Yang–Mills existence

Conditional estimates at source zero, complete current transport, and native target lifts

Status. The interacting four-dimensional Yang–Mills existence proof remains incomplete. In particular, this note does not prove a uniform bound on the full H susceptibility at source zero. It proves that bound for an extensive conditional fluctuation sector, estimates its actual generated ordinary coefficients and complete degree-two current packet, and retains the remaining conditional-mean covariance explicitly.

This note builds on [2] and the Wilson-source estimates [W], whose Chapter 161 fixes the inherited obligations. The original physical source, fine-mesh factors, conditional denominators and absolute scalar increments are preserved.

New finite-cutoff results:

- A genuine integration of a positive-density set of separated original links has uniformly extensive conditional variance *at source zero*. All its generated source coefficients have fixed-order local moment bounds.
- Original-coordinate marginals have quantitative effective-score jets. For the separated-link integration, the complete degree-two transport identity is bounded, including its ordinary covariance and current companion.
- A reference-centering component of an actual native target derivative has an explicit pinned-forest lift. Its complete normalized response is a controlled original sharp-boundary flux.
- Bare physical contact coefficients admit factorial bounds at every source order, with the original support of any one label retained.

The original nonlinear metric application is not established here: a primary paper [4] fixes constants for the linear propagator family, but the exterior/path convention and its nonlinear identification remain open. Complete S94 and the continuum hierarchy remain separate unfinished tasks.

The derivations are given in full within their stated scope. This is not external review or a formal proof certificate. The mass gap remains outside the target.

Contents

1 The fixed-source target and the inherited local input **3**

2 A complete conditional source calculation at zero **4**
2.1 An extensive set of conditionally independent links 4
2.2 Actual generated logarithmic coefficients 5
2.3 The resulting fixed-source susceptibility statement 6

3 Effective scores and inherited ordinary responses **6**
3.1 Density jets of any original-coordinate marginal 6
3.2 Local coefficient jets with the genuine denominator response 7
3.3 The cancellation of derivative losses for original current words 8

4 A complete degree-two current packet for this integration **8**
4.1 The pushed current is identified, including its score 8
4.2 All three terms of the conditional transport identity 8

5 What iteration would have to preserve **10**
5.1 An exact quadratic budget for complete positive transitions 10
5.2 An exact anisotropy identity does not supply the missing coercivity 10

6 An explicit native lift of reference-centering directions **11**
6.1 The exact sector being identified 11
6.2 The forest recursion and the logarithmic coordinate field 11
6.3 The normalized physical kernel is a centered boundary flux 12

7 Full derivatives, selectors and the scalar record **13**
7.1 The residual terms of an actual outer derivative 13
7.2 Ordered derivatives and physical source covariances 14
7.3 Absolute scalar increments and a restricted comparison 15

8 The native metric and parameter verification **15**
8.1 New primary evidence and its exact scope 15
8.2 The precise missing clauses 16
8.3 A weighted-operator alternative and its Schur-source obstacle 16

9 All-order factorial control of the bare contact sector **17**
9.1 An independent analytic remainder estimate 18

10 S94 and the continuum hierarchy: the remaining quantitative work **19**
10.1 A native wall tail is not yet an ultraviolet scalar sum 19
10.2 Obligation ledger 19

1 The fixed-source target and the inherited local input

Work with the original finite $SU(N)$ Wilson law, $N \geq 2$ fixed, on a four-dimensional periodic lattice of spacing a and n sites per direction:

$$\rho = e^{-\beta S_0}, \qquad S_0 = \sum_p s_p, \qquad s_p = N - \Re \operatorname{tr} U_p, \qquad d\mu_\beta = Z_\beta^{-1} \rho \, d\lambda_{\mathrm{Haar}}, \qquad \beta = c_\beta g^{-2}. \tag{1}$$

The positive constant c_β, group generators and original pointings are fixed. Physical test weights are field-independent and identical for every color component of a given bond. A coordinate index $i = (b, A)$ includes a complete original bond and a Lie-algebra component; X_i is the invariant Haar-skew-adjoint derivative. Coordinate projection below always retains or integrates *whole group variables*, never an arbitrarily selected component of a global logarithm chart.

Retain the reflection-compatible torus regime of [2]

$$\beta \geq 2, \qquad \frac{\log \beta}{n} \leq K, \tag{2}$$

with the fixed local chessboard and pointing stencils fitting on the torus. Its local moment theorem gives

$$\sup_p \mathbb{E} e^{t_0 \beta s_p} \leq A, \qquad \left\| \beta \sum_{p \in Y} s_p \right\|_{L^u(\mu_\beta)} \leq C u \, \#Y \quad (u \geq 1) \tag{3}$$

for a fixed $t_0 > 0$. These are moments in the original Wilson measure. They do not assert independence or a decay of connected covariances.

For clarity, the finite-size input behind (3) is

$$(c_- \beta^{-m/2})^{3n^4+1} \leq Z_\beta \leq (c_+ \beta^{-m/2})^{3n^4-4n^3+1}, \qquad m = N^2 - 1.$$

Disseminating a plaquette over n^4/v compatible cells bounds its exponential moment by a partition-function ratio raised to v/n^4. The unmatched surface power is $\beta^{2mv/n}$, controlled by (2). This is why that finite-size condition is retained.

Write

$$\begin{aligned} F_i &= X_i S_0 = \sum_{\mu \neq \nu_b} F_{i;\mu}, & P[f] &= \sum_i f_i F_i^2, \\ D[f] &= \sum_i f_i \sum_{\mu \neq \nu_b} F_{i;\mu}^2, & \ell[f] &= \sum_i f_i X_i^2 S_0, \qquad W_f = \sum_i f_i F_i X_i. \end{aligned} \tag{4}$$

The actual physical H packet remains

$$\begin{aligned} &O_f = \sigma G_f, \quad G_f = D[f] - P[f]/3, \quad \sigma = \beta a^{-2}/c_\beta, \\ &V_f = -\frac{a^{-2}}{3c_\beta} W_f, \qquad t_f = -\frac{\sigma}{3} P[f] + \frac{\sigma}{3\beta} \ell[f], \qquad r_f = \sigma D[f] - \frac{\sigma}{3\beta} \ell[f]. \end{aligned} \tag{5}$$

Thus $O_f = t_f + r_f$ and $D_{V_f} \rho = t_f \rho$, where $D_V u = -\operatorname{div}(uV)$. All fixed pointings are averaged in every member of the packet, with their actual shifted tests. The Casimir constant in ℓ is retained.

For the uniform real test, $G_H = G_1 \geq 0$ and $G_H \leq C_H S_0$. The exact susceptibility target is

$$\chi_{n,\beta} \equiv \frac{\beta^2}{n^4} \operatorname{Var}_{\mu_\beta}(G_H) = \left. \frac{d^2}{d\lambda^2} \frac{1}{n^4} \log \int e^{-\beta S_0 + \lambda \beta G_H} \, d\lambda_{\mathrm{Haar}} \right|_{\lambda = 0}. \tag{6}$$

The physical source parameter is $\eta = \lambda c_\beta a^2$. A uniform upper bound on (6) would give

$$\mathrm{Var}(O_H) \le C c_\beta^{-2} a^{-4} n^4 = C T^4 a^{-8}, \qquad T = na,$$

matching the order of the nonlinear lower bound in [W], PD12.55. The real-source averaged estimate of [2] does not prove this value-at-zero bound. The calculations below do not use that invalid implication.

2 A complete conditional source calculation at zero

2.1 An extensive set of conditionally independent links

Expand the literal finite physical average as

$$G_f = \sum_\alpha f_\alpha h_\alpha(U). \tag{7}$$

Here α retains its original bond anchor, pointing and average weight; h_α is a nonnegative directional-force-square expression. It has a fixed finite determining link set T_α, and

$$0 \le h_\alpha \le C \sum_{p \in Y_\alpha} s_p.$$

Use a finite-range graph on original links, joining two when they occur in one plaquette or together in a complete source/current stencil. Its degree is uniformly bounded. A graph coloring gives a fixed number Q of independent classes, one of which, denoted $\mathcal{S}$, has at least $\#E/Q$ links. Any one such class may be used.

Let X contain all retained links $E \setminus \mathcal{S}$. No plaquette or complete marked stencil contains two links in $\mathcal{S}$. Consequently

$$\begin{aligned} S_0 &= S_{\mathrm{ext}}(X) + \sum_{e \in \mathcal{S}} S_e(U_e; X), \\ G_{f_j} &= G_{0,j}(X) + \sum_{e \in \mathcal{S}} G_{e,j}(U_e; X). \end{aligned} \tag{8}$$

S_e is the sum of original plaquette actions containing e. Assign a marked summand to the unique selected link in its determining set, if there is one, and otherwise to $G_{0,j}$. An assigned summand may include an exterior-only part; this causes no error in the exact equality.

Define the genuine positive conditional factors

$$z_e(X) = \int e^{-\beta S_e(U_e;X)}\, dU_e, \qquad K_e(dU_e \mid X) = z_e(X)^{-1} e^{-\beta S_e(U_e;X)}\, dU_e. \tag{9}$$

The original conditional law is $K_{\mathcal{S}} = \prod_e K_e$, and the unnormalized retained density is

$$\rho_X(X) = e^{-\beta S_{\mathrm{ext}}(X)} \prod_e z_e(X), \qquad \int \rho_X\, dX = Z_\beta. \tag{10}$$

This is an exact integration of the original physical law over all selected group variables. There is no artificial field cutoff, discarded bad-field set or substituted denominator. It is a useful exact reorganization of the path integral; its identification with a particular selected Bałaban operation is not asserted.

For each e, retain its fixed complete plaquette record Y_e, containing the local actions and all assigned marked factors, and put

$$E_e = \beta \sum_{p \in Y_e} s_p, \qquad A_j(e) = \sum_{\alpha \text{ assigned to } e} |f_{j,\alpha}|, \qquad O_{e,j} = \sigma G_{e,j}. \tag{11}$$

The original averaging weights occur once in A_j. Finite incidence and (3) give

$$|O_{e,j}| \le C a^{-2} A_j(e) E_e, \qquad \|O_{e,j}\|_{L^u(\mu_\beta)} \le C u\, a^{-2} A_j(e). \tag{12}$$

These are global-law moments, not pointwise uniform conditional moments for every exterior field.

2.2 Actual generated logarithmic coefficients

Use the unchanged bare source $\sum_j \eta_j O_{f_j}$ with distinct commuting squarefree labels. If

$$D_e(\eta; X) = \int e^{-\beta S_e + \sum_j \eta_j O_{e,j}}\, dU_e,$$

then exactly

$$K_{\mathcal{S}} e^{\sum_j \eta_j O_{f_j}} = e^{\sum_j \eta_j \sigma G_{0,j}} \prod_{e \in \mathcal{S}} \frac{D_e(\eta; X)}{z_e(X)}. \tag{13}$$

The displayed quotient uses the actual source-zero z_e. For a label set I of size $r \ge 2$, the logarithm has coefficient

$$A_I^{\mathrm{out}}(X) = \sum_{e \in \mathcal{S}} \kappa_{K_e}(O_{e,j} : j \in I). \tag{14}$$

This is a generated ordinary connected coefficient, not just a transported input contact.

Theorem 2.1 (Actual conditional cumulants)**.** *Define the original overlap count*

$$\mathcal{N}_I^{\mathcal{S}} = \sum_{e \in \mathcal{S}} \prod_{j \in I} A_j(e).$$

For every fixed $r \ge 2$ and $u \ge 1$,

$$\boxed{\left\| A_I^{\mathrm{out}} \right\|_{L^u(\mu_X)} \le C_r u^r a^{-2r} \mathcal{N}_I^{\mathcal{S}}.} \tag{15}$$

Its raw absolute mass is bounded by $Z_\beta C_r a^{-2r} \mathcal{N}_I^{\mathcal{S}}$. An independent factor e^s multiplies absolute mass by $e^{\Re s}$.

Proof. For a partition π of I, its cumulant term has coefficient $(|\pi| - 1)!$ and a product of conditional moments. By (12), its absolute value is bounded by

$$C^r a^{-2r} \prod_j A_j(e) \prod_{B \in \pi} K_e(E_e^{|B|}).$$

Conditional Hölder gives $\prod_{B \in \pi} K_e(E_e^{|B|}) \le K_e(E_e^r)$. After summing the finite partition formula, conditional Jensen and (3) give

$$\|\kappa_{K_e}\|_{L^u(\mu_X)} \le C_r a^{-2r} \prod_j A_j(e)\, \|E_e\|_{L^{ru}(\mu_\beta)}^r \le C_r u^r a^{-2r} \prod_j A_j(e).$$

Minkowski sums over e. Equation (10) supplies the original raw normalization. □

The count is linear in original support volume for bounded tests. It vanishes unless one selected-link neighborhood touches every source label. It does not replace a by a later mesh. A safe explicit bound on the order dependence is $C_r \le C^r (r!)^2$: the partition coefficient sum has exponential generating function $-\log(2-e^z)$, hence is at most $C^r r!$, and the local energy moment supplies another $C^r r!$. This is a Gevrey-two bound, not a uniform analytic radius for every conditional logarithm.

2.3 The resulting fixed-source susceptibility statement

Theorem 2.2 (Extensive conditional variance at zero)**.** *For the uniform real H source, the exact identity is*

$$\mathrm{Var}_{\mu_\beta}(G_H) = \sum_{e\in\mathcal{S}} \mathbb{E}\,\mathrm{Var}_{K_e}(G_e) + \mathrm{Var}_{\mu_X}(K_{\mathcal{S}} G_H). \tag{16}$$

The first term satisfies

$$0 \le \sum_e \mathbb{E}\,\mathrm{Var}_{K_e}(G_e) \le C\beta^{-2}|\mathcal{S}|, \qquad \mathbb{E}\,\mathrm{Var}(O_H \mid X) \le C a^{-4}|\mathcal{S}|. \tag{17}$$

Thus $\chi_{n,\beta} = \chi_{\mathrm{fast}} + \chi_{\mathrm{retained}}$, with

$$0 \le \chi_{\mathrm{fast}} \le C, \qquad \chi_{\mathrm{retained}} = \frac{\beta^2}{n^4}\,\mathrm{Var}_{\mu_X}(K_{\mathcal{S}} G_H). \tag{18}$$

Proof. Conditional independence makes the conditional variance of (8) the sum of the single-link conditional variances. The law of total variance gives (16). For each term, $\mathbb{E}\,\mathrm{Var}_{K_e}(G_e) \le \mathbb{E}G_e^2 \le C\beta^{-2}$ by (12). Sum and restore σ^2 for the physical mark. □

This is a justified additional mechanism at *the original source value zero*: conditional independence plus local Wilson moments bounds an extensive set of actual fluctuations. It does not control the retained covariance in (18). That retained first source is an exact sum of local conditional means and unintegrated terms, whose global connected correlations still need an estimate.

3 Effective scores and inherited ordinary responses

3.1 Density jets of any original-coordinate marginal

The following result is not limited to separated links. Let X be any retained set of whole original links and Y the complement, with

$$\rho_X(X) = \int \rho(X,Y)\,dY, \qquad K(F) = \rho_X^{-1} \int \rho F\,dY.$$

For an ordered word $J = (j_1,\dots,j_k)$ of retained invariant derivatives, set $X_J = X_{j_k}\cdots X_{j_1}$.

Theorem 3.1 (Marginal density and effective-score jets)**.** *For every fixed $k \ge 1$ and $u \ge 1$,*

$$\frac{X_J \rho_X}{\rho_X} = K(P_J), \qquad P_J = \rho^{-1} X_J \rho, \tag{19}$$
$$\|X_J \rho_X/\rho_X\|_{L^u(\mu_X)} + \|X_J \log \rho_X\|_{L^u(\mu_X)} \le C_k (\beta u)^{k/2}.$$

No reciprocal lower bound on ρ_X is used.

Proof. Differentiate under the compact integral to obtain the exact first identity. The product rule expands P_J into products of factors $\beta X_B S_0$ for nonempty ordered subwords, with total derivative degree k. A singleton satisfies $\|\beta F_i\|_{L^u} \le C\sqrt{\beta u}$ by the first-force bound and (3). Every fixed action derivative of order $l \ge 2$ is bounded pointwise, with only the original finite plaquette incidence; hence $|\beta X_B S_0| \le C_l \beta \le C_l \beta^{l/2}$. Hölder bounds each product by $C_k(\beta u)^{k/2}$. Conditional Jensen gives the density-jet bound. The derivatives of $\log \rho_X$ are finite polynomials in normalized density jets of the same total degree; a further Hölder inequality proves their bound. No invariant derivatives are commuted. □

In particular, the actual effective force is

$$\bar{F}_i = K(F_i) = -\beta^{-1} X_i \log \rho_X, \qquad X_j \bar{F}_i = K(X_j F_i) - \beta \operatorname{Cov}_K(F_i, F_j). \tag{20}$$

The covariance is part of the effective Hessian. It cannot be omitted. The bounds are

$$\left\|\bar{F}_i\right\|_{L^u} \le C\beta^{-1/2}\sqrt{u}, \qquad \left\|X_J \bar{F}_i\right\|_{L^u} \le C_k \beta^{(k-1)/2} u^{(k+1)/2}. \tag{21}$$

Normalized directions gX_i obey $g\sqrt{\beta} = \sqrt{c_\beta}$, so their fixed-order score jets, and the corresponding normalized jets of $\bar{F}_i/g$, have no coupling loss.

These are individual-coordinate bounds. For arbitrary marginals, they do not give decay between distant retained coordinates or a uniform norm for an arbitrary sum of coordinate contractions. Such sums keep their actual counts.

3.2 Local coefficient jets with the genuine denominator response

Return to the separated-link step and set $a_{I,e} = \kappa_{K_e}(O_{e,j} : j \in I)$.

Proposition 3.2 (Conditional source jets)**.** *For a fixed source order r and derivative order k,*

$$\|X_J a_{I,e}\|_{L^u(\mu_X)} \le C_{r,k} a^{-2r} \beta^{k/2} u^{r+k/2} \prod_{j \in I} A_j(e). \tag{22}$$

Every derivative outside the actual finite output determining record of e vanishes.

Proof. A local physical H term starts as βa^{-2} times two first-force factors. One derivative can remove at most one such factor; two or more can remove both. Compact bounds on the remaining action derivatives therefore give

$$\|X_J O_{e,j}\|_{L^u} \le C_k a^{-2} \beta^{k/2} u A_j(e).$$

For a product of r such terms, derivatives of the source factors and derivatives of the original density have total order k. Hölder and Theorem 3.1 give the displayed $\beta^{k/2} u^{r+k/2}$ bound on each normalized numerator jet.

For $N = \int \rho F \, dY$ and $D = \rho_X$, differentiate $D(KF) = N$. For the ordered word J, the exact Leibniz identity is

$$DX_J(KF) = X_J N - \sum_{\emptyset \ne A \subseteq [k]} (X_A D) X_{A^c}(KF). \tag{23}$$

Every subword preserves its original relative order. Induction on k, using the true normalized density jets in (19), proves the same bound for conditional moment derivatives. Differentiating the finite cumulant partition formula proves (22). The local action, source factors and z_e fix its output record, proving the support assertion. □

The same proof covers the actual local r_H and t_H terms. Their force squares have identical bounds; their full Casimir terms have size $O(a^{-2})$ and fixed-order compact derivatives. No scalar partner is dropped.

3.3 The cancellation of derivative losses for original current words

Pull $a_{I,e}(X)$ back to the full original configuration. An ordered word of k bare directions $W_{h_l} = \sum_i h_{l,i} F_i X_i$, with field-independent weights, expands as

$$W_{h_k} \cdots W_{h_1} F = \sum_{d=1}^{k} \sum_{|J|=d} c_J(U) X_J F. \tag{24}$$

Each monomial in c_J contains at least d first-force factors. Indeed, applying the next W adds a force. If it differentiates a coefficient, it consumes at most one old first-force factor and leaves d unchanged. If it differentiates the remaining test, it raises d and the force count by one. This is an ordered induction, with no commutator rearrangement.

On a fixed local record the coordinate choices and its enlargement are fixed-order counts, so

$$\|c_J\|_{L^u} \le C_k \beta^{-d/2} u^{d/2} \prod_l \|h_l\|_\infty .$$

Combining with (22) cancels the $\beta^{d/2}$ derivative loss:

$$\|W_{h_k} \cdots W_{h_1} (a_{I,e} \circ q)\|_{L^u(\mu_\beta)} \le C_{r,k} a^{-2r} u^{r+k} \prod_j A_j(e) \prod_l \|h_l\|_\infty . \tag{25}$$

The actual physical fields V_{h_l} contribute their additional original factor $(3c_\beta)^{-k} a^{-2k}$. This is not replaced by a current-scale mesh. Internal-field-dependent weights require their own derivatives and are not covered by pretending those weights are numerical.

4 A complete degree-two current packet for this integration

4.1 The pushed current is identified, including its score

For the actual coordinate projection q and $\nu = \mu_X$, the pushed vector field is

$$\bar{V}_f = K(dq V_f) = -\frac{a^{-2}}{3c_\beta} \sum_{i \text{ retained}} f_i \bar{F}_i X_i. \tag{26}$$

Testing against a retained function and integrating the internal compact group variables gives

$$q_*(D_{V_f} \mu_\beta) = D_{\bar{V}_f} \nu, \qquad \bar{t}_f = K(t_f) = \rho_X^{-1} D_{\bar{V}_f} \rho_X = \frac{a^{-2}}{3c_\beta} \sum_{i \text{ retained}} f_i (X_i \bar{F}_i - \beta \bar{F}_i^2). \tag{27}$$

Using (20) verifies the last identity directly. For an internal coordinate, its original score integrates to zero on its complete compact fiber. For a retained coordinate, the effective-force covariance supplies precisely the missing denominator response.

For separated links, $\rho_X = e^{-\beta S_{\text{ext}}} \prod_e z_e$ has local factors. Hence $\bar{F}_i$ has a bounded enlarged local determining stencil. This locality does not extend automatically to an arbitrary coordinate marginal.

4.2 All three terms of the conditional transport identity

Let b_{12} be the complete bare compensated contact from [2]:

$$b_{12} = \tfrac{1}{2}(V_1 t_2 + V_2 t_1) + V_1 r_2 + V_2 r_1. \tag{28}$$

The actual output logarithmic coefficient is $A_{12}^{\mathrm{out}} = \mathrm{Cov}_K(O_1, O_2)$. Set $\bar{r}_i = K(r_i)$, $v_i = dqV_i$ and define its compensated counterpart with the actual pushed current:

$$\widehat{b}_{12} = A_{12}^{\mathrm{out}} + \tfrac{1}{2}(\bar{V}_1\bar{t}_2 + \bar{V}_2\bar{t}_1) + \bar{V}_1\bar{r}_2 + \bar{V}_2\bar{r}_1.$$

The exact conditional transport identity of [W] is

$$\begin{aligned} \widehat{b}_{12}\nu &= K(b_{12})\nu + \mathrm{Cov}_K(r_1, r_2)\nu + D(J_{12}\nu), \\ J_{12} &= \tfrac{1}{2}\{\mathrm{Cov}_K(t_2, v_1) + \mathrm{Cov}_K(t_1, v_2)\} + \mathrm{Cov}_K(r_2, v_1) + \mathrm{Cov}_K(r_1, v_2). \end{aligned} \tag{29}$$

For example, the underlying product identity is

$$\mathrm{Cov}_K(t_1, r_2)\nu = \{K(V_1 r_2) - \bar{V}_1\bar{r}_2\}\nu + D(\mathrm{Cov}_K(r_2, v_1)\nu).$$

It follows from $t_1 r_2\mu = D_{V_1}(r_2\mu) + (V_1 r_2)\mu$. Symmetrizing its current-current version gives (29). This check fixes its signs and factors.

For the selected separated links, all conditional covariances decompose into terms using one K_e. Different selected links are conditionally independent, and exterior-only pieces have zero conditional covariance. Enlarge Y_e once to include both source/current anchors and every output component reached by the current. Define $A_i^{\mathrm{comp}}(e)$ by the actual sum of absolute test weights on this completion, and

$$\mathcal{N}_{12}^{\mathcal{S},\mathrm{comp}} = \sum_e A_1^{\mathrm{comp}}(e) A_2^{\mathrm{comp}}(e).$$

It is the actual extended overlap count; it is not squared merely because the two source labels differ.

Theorem 4.1 (Complete conditional degree-two estimate)**.** *For this all-fields separated-link integration,*

$$\boxed{\left\|\widehat{b}_{12}\right\|_{L^1(\mu_X)} \le Ca^{-4}(\mathcal{N}_{12}^{\mathrm{bare}} + \mathcal{N}_{12}^{\mathcal{S},\mathrm{comp}}).} \tag{30}$$

$\mathcal{N}_{12}^{\mathrm{bare}}$ *is the original complete bare overlap count in [2]. Raw absolute mass multiplies* (30) *by* Z_β *and any independent* $e^{\Re s}$*.*

Proof. Conditional Jensen and the contact theorem of [2] bound $K(b_{12})$. Theorem 2.1, applied to the local ordinary pieces, bounds $\mathrm{Cov}_K(r_1, r_2)$ by $Ca^{-4}\mathcal{N}_{12}^{\mathcal{S},\mathrm{comp}}$ in L^1.

A local component of J_{12} contains a physical packet of size a^{-2} and a current coefficient of size $a^{-2}\beta^{-1/2}$. The source-jet argument gives

$$\begin{aligned} \|J_{12,e}\|_{L^u} &\le Ca^{-4}\beta^{-1/2}u^{3/2}A_1^{\mathrm{comp}}(e)A_2^{\mathrm{comp}}(e), \\ \|X_j J_{12,e}\|_{L^u} &\le Ca^{-4}u^2 A_1^{\mathrm{comp}}(e)A_2^{\mathrm{comp}}(e). \end{aligned} \tag{31}$$

The Haar-invariant basis is divergence-free, so

$$\rho_X^{-1} D(J\rho_X) = -\sum_j \{X_j J_j + J_j X_j \log \rho_X\}. \tag{32}$$

The score norm from (19) is $O(\sqrt{\beta u})$. Hölder cancels the inverse square root of β in (31). Each e has only a bounded number of actual output coordinates; summing them and then e proves the current-row estimate and (30). □

This proves a genuine complete conditional transition estimate. It does not bound the subsequent global ordinary covariance $\mathrm{Var}_\nu(\bar{r}_H)$. A single-fiber conditional covariance in (29) and that retained covariance are different quantities.

5 What iteration would have to preserve

5.1 An exact quadratic budget for complete positive transitions

Let $X_0 \to X_1 \to \cdots \to X_K$ be a verified Markov realization of complete positive source-free mass-preserving transitions, under the original source-zero probability. All selectors needed for that complete realization are included. Let the real L^2 physical marks O_i at birth be measurable with respect to X_0, so that $m_i^0 = O_i$ and $C_{ij}^0 = 0$. Define

$$m_i^k = \mathbb{E}(O_i \mid X_k), \qquad C_{ij}^k = \mathrm{Cov}(O_i, O_j \mid X_k), \qquad D_{ij}^k = \mathrm{Cov}(m_i^k, m_j^k \mid X_{k+1}).$$

The exact logarithmic recursion is

$$m_i^{k+1} = \mathbb{E}(m_i^k \mid X_{k+1}), \qquad C_{ij}^{k+1} = \mathbb{E}(C_{ij}^k \mid X_{k+1}) + D_{ij}^k. \tag{33}$$

The Markov property justifies removing future variables after conditioning on X_k; no nested-coordinate assumption is necessary.

Proposition 5.1 (One ordinary covariance budget)**.** *Every matrix D^k is positive semidefinite, and*

$$\mathrm{Var}(O_i) = \sum_{k=0}^{K-1} \mathbb{E} D_{ii}^k + \mathrm{Var}(m_i^K), \tag{34}$$

$$\sum_{k=0}^{K-1} \mathbb{E}|D_{ij}^k| \le \sqrt{\mathrm{Var}(O_i)\,\mathrm{Var}(O_j)}.$$

Proof. The law of total variance makes $\mathbb{E} D_{ii}^k = \mathrm{Var}(m_i^k) - \mathrm{Var}(m_i^{k+1})$, which telescopes. Conditional Cauchy–Schwarz, then Cauchy–Schwarz in probability and in k, gives the second inequality. □

This prevents charging the same complete ordinary variance separately for every intermediate history. It supplies no new upper bound on the initial variance itself. Incoming mixed contacts and scalar counterterms are transported separately and are not automatically positive members of this budget. Current-eliminated representations must be recombined with their kernel responses before they are identified with the unchanged physical source.

The separated-link theorem bounds the first term of (34) at source zero. Iterating its proof without controlling the newly generated determining sets would be invalid. After elimination, effective scores and source carriers can acquire growing range. A graded contraction or a summable spatial connected estimate must control this growth. This is the remaining MT36 issue, now expressed in an exact conditional recursion.

5.2 An exact anisotropy identity does not supply the missing coercivity

There is a useful special structure of the uniform H source. If $S_{\mu\nu}$ denotes the action in one unoriented plane, define

$$P_1 = S_{12} + S_{34}, \quad P_2 = S_{13} + S_{24}, \quad P_3 = S_{14} + S_{23}, \qquad S_0 = P_1 + P_2 + P_3,$$

and

$$A_1 = (P_1 - P_2)/\sqrt{2}, \qquad A_2 = (P_1 + P_2 - 2P_3)/\sqrt{6}.$$

At each bond the three P_j contribute its three directional forces. Hence exactly

$$G_H = \sum_{j=1}^{3} |\nabla(P_j - S_0/3)|^2 = |\nabla A_1|^2 + |\nabla A_2|^2. \tag{35}$$

The Haar Casimir gives $\Delta A_\alpha = -4C_F A_\alpha$. For the Wilson generator $\mathcal{L} = \Delta - \beta \nabla S_0 \cdot \nabla$,

$$G_H = \tfrac{1}{2}\mathcal{L}(A_1^2 + A_2^2) - \sum_{\alpha=1}^{2} A_\alpha \mathcal{L} A_\alpha. \tag{36}$$

The last term contains products of extensive anisotropies and is not a local contact bound. Thus this Ward reformulation does not establish coercivity of the retained ordinary sector. The strong-coupling functional inequalities in [7] concern a small inverse-coupling interval and cannot be imported onto the $\beta \to \infty$ trajectory considered here.

6 An explicit native lift of reference-centering directions

6.1 The exact sector being identified

Use the admitted R-II reference denominator and its fixed-target minimum from [W], NB12 and RV11; the original source is [3]. Its relative target has the literal form

$$W = CV_\Lambda, \qquad C_e = e^{gB_e},$$

using skew-Hermitian generators. Its original pinned forest, exterior fields, constraint rows, spatial weight and boundary frames are fixed. A variation of the interior reference center is a component of an outer physical derivative when that derivative moves the live reference. It is not generally the entire original fine-link current.

Let the dynamic graph have edges E, vertices V and pinned vertices P, and choose the original allowed forest $\mathcal{T}$ with one pin per tree. The actual free group and Lie-coordinate counts are

$$q = |E| - |V| + |P|, \qquad n_{\mathrm{nat}} = mq, \qquad m = \dim SU(N). \tag{37}$$

All prescribed exterior data must be fixed by the pinned gauge action; otherwise its response belongs to the residual terms below. Gauge covariance and uniqueness refer to the admitted fixed-target minimum with its full record, not to a newly asserted global branch.

6.2 The forest recursion and the logarithmic coordinate field

Fix g and let $\dot{V}_e = a_e V_e$, so $a_e = \dot{V}_e V_e^{-1}$. Initially a may depend on the retained target but not on C. Orient each tree edge from root to child for this calculation. The pinned gauge $h(t)$ restoring its old tree value obeys

$$h_x(t) V_e(t) h_y(t)^{-1} = V_e(0), \qquad h_p(t) = 1.$$

Its velocity $\xi_x = \dot{h}_x(0)$ is therefore

$$\xi_p = 0, \qquad \xi_y = \mathrm{Ad}_{V_e^{-1}}(\xi_x + a_e), \qquad |\xi_x| \le \sum_{t \in \mathrm{path}_{\mathcal{T}}(p,x)} |a_t|. \tag{38}$$

Changing the orientation of an original edge uses its actual reversed left velocity. This is a calculation convention, not a new forest.

On a free edge $e = (x, y)$, the changed physical target in the original section is

$$C'_e(t) = h_x(t) C_e V_e(t) h_y(t)^{-1} V_e(0)^{-1}.$$

It gives the explicit internal field

$$v_a C_e = \xi_x C_e + C_e(a_e - \mathrm{Ad}_{V_e}\, \xi_y). \tag{39}$$

This field represents target differentiation internally. The simultaneous compensator that keeps the physical orbit fixed has the opposite sign.

The coefficients are independent of the free C variables. Each term is a left or right invariant field, so

$$\mathrm{div}_{\mathrm{Haar}}\, v_a = 0. \tag{40}$$

For a gauge-invariant factor $L(W; b)$ depending on this center only through W, with all other data b fixed,

$$X_a L(CV; b) = v_a L(CV; b). \tag{41}$$

Indeed, $v_a W - X_a W = \xi_x W - W \xi_y$, whose allowed gauge derivative annihilates L.

Define $\mathrm{dexp}_z = (1 - e^{-\mathrm{ad}_z})/\,\mathrm{ad}_z$ with its removable value at zero. In the actual logarithmic coordinates,

$$v^B_{a,e}(B) = g^{-1}\, \mathrm{dexp}^{-1}_{gB_e}\, (\,\mathrm{Ad}_{C_e^{-1}}\, \xi_x + a_e - \mathrm{Ad}_{V_e}\, \xi_y). \tag{42}$$

On the fixed small logarithm chart containing the original cutoff, every fixed internal order satisfies

$$|\partial^\alpha_B v^B_{a,e}| \le C_r g^{|\alpha|-1} \mathcal{A}_e(a), \qquad \mathcal{A}_e(a) = |\xi_x| + |a_e| + |\xi_y|, \qquad |\alpha| \le r. \tag{43}$$

Derivatives in a different free edge vanish. Analyticity of inverse dexp and adjoint maps on the fixed group chart proves the estimate, with one g for each B derivative.

Retain the original incidence sum

$$\mathcal{A}_{\mathcal{T}}(a) = \sum_{e\ \text{free}} \mathcal{A}_e(a) \le \sum_{e\ \text{free}} |a_e| + \sum_{t \in \mathcal{T}} N_t |a_t|. \tag{44}$$

N_t counts free-edge endpoints whose root path contains t. For maximum path length $l_{\mathcal{T}}$, this is at most $q(2l_{\mathcal{T}} + 1)\, \|a\|_\infty$. These counts remain part of the estimate. If a depends on internal fields, its derivatives and its Haar divergence must be restored; (40) no longer follows by this argument.

6.3 The normalized physical kernel is a centered boundary flux

Let $j_g(B)dB$ be the actual quotient Haar measure in the free coordinates, retaining its $g^{n_{\mathrm{nat}}}$ factor if not already extracted. Put

$$f_{\mathrm{raw}}(B; V, b) = j_g(B) L(CV; b), \qquad D(V, b) = \int_{\Omega_R} f_{\mathrm{raw}}\, dB, \qquad K = \mathbf{1}_{\Omega_R} f_{\mathrm{raw}}\, dB/D, \tag{45}$$

where $\Omega_R = \{|B_e|_* < R \text{ for every free } e\}$ is the unchanged native product-norm cutoff, $R = M_0 p_0(g)$. For a carrier F, write $N_F = \int_{\Omega_R} F f_{\mathrm{raw}}\, dB$, so $KF = N_F/D$. Haar divergence-freeness means

$$\mathrm{div}_B\, v^B_a + v^B_a \log j_g = 0, \qquad X_a f_{\mathrm{raw}} = \mathrm{div}_B(f_{\mathrm{raw}} v^B_a).$$

The flat-coordinate divergence alone is not zero.

For a complete physical carrier $F(CV;b)$ satisfying the same record condition, the divergence theorem gives

$$X_a \log D = D^{-1} \int_{\partial\Omega_R} f_{\mathrm{raw}}(v_a^B \cdot n)\, d\Sigma, \tag{46}$$
$$X_a(KF) = D^{-1} \int_{\partial\Omega_R} (F - KF) f_{\mathrm{raw}}(v_a^B \cdot n)\, d\Sigma.$$

The full denominator derivative is present. The ordinary product cutoff is Lipschitz; its corners have zero surface measure. These are single boundary fluxes, not products of delta distributions.

Use [W], NB12.6–12 on this admitted reference: its exact extracted density has a Gaussian upper and lower envelope, fixed-order internal jets bounded by $C_{M,r}(1+|B|)^r f(B)$, and the original boundary trace

$$\widehat{D}^{-1} \int_{\partial\Omega_R} (1+|B|)^p e^{\theta|B|^2} f(B)\, d\Sigma \le C_{M,p,\theta} q R^{m-1}(1+R)^p e^{-c_M R^2}. \tag{47}$$

Here $n_{\mathrm{nat}} = mq$ is retained in the native constants, the admitted component is contained in its original fixed current-scale cube, and θ uses the already available Gaussian margin.

Proposition 6.1 (Native target-centering response)**.** *If $|F(B)| \le A_F(1+|B|)^p e^{\theta|B|^2}$ on this reference, then*

$$|X_a \log D| \le C_M g^{-1} \mathcal{A}_{\mathcal{T}}(a) R^{m-1} e^{-c_M R^2}, \tag{48}$$
$$|X_a(KF)| \le C_{M,p,\theta} A_F g^{-1} \mathcal{A}_{\mathcal{T}}(a) R^{m-1}(1+R)^p e^{-c_M R^2}.$$

Proof. On a face belonging to edge e, only its vector block contributes to $v \cdot n$. Apply the proof of (47) face by face and use $|v_{a,e}^B| \le C g^{-1} \mathcal{A}_e(a)$. Summing gives (44); there is no additional q to insert a second time. The native bulk moment bounds KF in the centered term of (46). The common raw prefactor cancels only in these displayed ratios; raw boundary masses still equal D times the corresponding bounds. □

For a current-normalized direction gX_a, the displayed g^{-1} cancels; a direction $g_j X_a$ with an older saved coupling retains g_j/g. Every old mesh factor and saved numerical coefficient belongs to A_F or to the actual velocity. The current-coupling wall tail does not erase a separately saved inverse power of an older mesh.

7 Full derivatives, selectors and the scalar record

7.1 The residual terms of an actual outer derivative

Let X now be an actual retained-variable derivative at fixed g, and let v be only its identified target-centering lift. Retain the literal raw density f, complete selector χ, observation q and carrier F, with $D = \int \chi f$. The normalized formulas below require $f \neq 0$ on the working chart and $D \neq 0$; both hold for the positive admitted native fiber. Zero-mass selected histories remain in their unnormalized tested realization. Define

$$H_X = X - v, \qquad S_X = \frac{Xf - \mathrm{div}_B(fv)}{f}, \qquad \mathcal{B}_X = D^{-1} f(X\chi - v\chi). \tag{49}$$

The last expression is a distribution acting on the original tested fiber. No logarithm of a sharp selector is taken. A selected term with no established differentiable realization must remain in its unnormalized tested-kernel form.

Differentiation and integration by parts give exactly

$$X \log D = K S_X + \mathcal{B}_X(1), \tag{50}$$
$$X(KF) = K(H_X F) + \mathrm{Cov}_K(F, S_X) + \mathcal{B}_X(F) - (KF)\mathcal{B}_X(1).$$

For fixed $\chi = \mathbf{1}_{\Omega_R}$, $-v\chi = (v \cdot n)d\Sigma$, reproducing (46). Moving or additional selectors keep $X\chi - v\chi$; their support is not automatically the native Gaussian wall. If $f = j_g L$ and the fixed-target Haar divergence is zero, $S_X = (X - v) \log L$. An internal-field-dependent velocity adds its actual divergence and coefficient responses.

For the pushed observation measure $\nu = q_* K$, the tested form is

$$X\nu = -\operatorname{div}_y q_*[(H_X q)K] + q_*[(S_X - KS_X)K] + q_*\mathcal{B}_X - \mathcal{B}_X(1)\nu. \tag{51}$$

The four terms respectively retain the observation current, the ordinary effective score and the complete centered selector response. A separately retained live-reference coordinate or independent current slot can make $H_X q \neq 0$ even when a physical W-only test has zero drift. This distinction is essential for the actual mixed histories.

7.2 Ordered derivatives and physical source covariances

For a simple product with constant Lie directions,

$$X_{a_r}^V \cdots X_{a_1}^V L(CV) = R_{a_1}^C \cdots R_{a_r}^C L(CV),$$

where R_a^C differentiates $C \mapsto Ce^{ta}$. The internal word reverses order. In the actual forest, differentiating target parameters also differentiates $\xi(V, a)$ and its adjoint factors. Each such coefficient derivative remains Haar-divergence-free when its coefficients remain independent of C and g is fixed. Induction represents an ordered pure-target derivative as a finite sum of nonempty ordered internal words, with all coefficient derivatives retained.

Integrating the leftmost internal field once produces one original boundary flux; remaining internal derivatives act on the actual density and carrier. If the carrier's internal jets through order $r - 1$ obey their stated envelopes, NB12 and (43) yield

$$D^{-1}|X_w N_F| \leq C_{M,r,p} A_F \mathcal{A}_w g^{-r} R^{m-1}(1 + R)^{p+r-1} e^{-c_M R^2}. \tag{52}$$

$\mathcal{A}_w$ is the actual finite sum of forest incidences and target-coefficient jets generated by that word, not a universal one. Normalized derivatives use the ordered quotient rule (23) with the genuine D, retaining every denominator derivative. The conclusion is at each fixed order, without an asserted all-order growth bound for this native sector.

There is also a useful exact finite-source identity. For the unchanged source $A_\eta = \sum_i \eta_i O_i(W)$, let $G_\eta = Ke^{A_\eta}$ and define the outward boundary functional $\mathcal{B}_a$ using (46). Then

$$X_a \log G_\eta = \mathcal{B}_a(e^{A_\eta}/G_\eta) - \mathcal{B}_a(1). \tag{53}$$

At degree two its boundary carrier is exactly

$$(O_1 - KO_1)(O_2 - KO_2) - \mathrm{Cov}_K(O_1, O_2). \tag{54}$$

An existing mixed coefficient adds $A_{12} - KA_{12}$. Thus a reference-centering *derivative* of the covariance can have a small wall bound when this actual carrier has the required jets. The covariance itself has not been canceled or bounded by normalization.

7.3 Absolute scalar increments and a restricted comparison

Retain the original extraction

$$D = P\widehat{D}, \qquad I = \log \widehat{D}, \qquad \log P = -g^{-2}A_0 + c_{\mathrm{coord}}. \tag{55}$$

The coordinate scalar includes the actual $n_{\mathrm{nat}} \log g$ and Haar normalization. At a fixed graph and g, those particular coordinate terms have zero target derivative, but their absolute values remain in the mass. Equations (50) and (55) give

$$\begin{aligned} XI &= KS_X + \mathcal{B}_X(1) + g^{-2}XA_0 - Xc_{\mathrm{coord}}, \\ X\log(e^s D) &= Xs + KS_X + \mathcal{B}_X(1). \end{aligned} \tag{56}$$

An independent Xs is not bounded by a normalized-kernel identity. The literal numerator partner has the exact cancellation

$$e^s \frac{e^{-g^{-2}A_0}N}{D} = e^{s - c_{\mathrm{coord}} - I}N. \tag{57}$$

Every factor on the right remains part of the absolute scalar and carrying record.

For a path $V(t)$ of reference centers on one identical full physical record, assume its whole path remains in the admitted branch with uniform native bounds. Integrating (48) gives

$$\left|\log \frac{D_1}{D_0}\right| \le C_M g^{-1} R^{m-1} e^{-c_M R^2} \int_0^1 \mathcal{A}_{\mathcal{T}}(a(t))\, dt. \tag{58}$$

The complete carrier has the corresponding bound with its actual envelope. The endpoints use their original denominators and cutoffs. This is a restricted reference-center comparison; differing spatial actions, exterior maps, records, wrapped components or terminal terms require additional comparisons.

8 The native metric and parameter verification

8.1 New primary evidence and its exact scope

The original background-propagator paper [4] gives the following verified clauses. Theorem 3.1, p. 397, fixes M_1, δ_0, a_0, B_0 depending only on d, L, requires $M \ge M_1$ and $M\alpha_0 \le a_0$, and separates the extra Hölder-exponent dependence. Page 399 states independence from the domain sequence under the original geometric conditions. Theorem 3.12 and its following paragraph, pp. 423–424, give common constants for the corrected propagators and constrained linear operators, under both regularity conditions (3.35), (3.36) and the required sufficiently small α_0. Its distance is explicitly referred to PRTII (2.36). The paper also uses an outer domain with Dirichlet boundary conditions; one cannot identify every Ω_0 with the whole torus.

This verifies a fixed-parameter-first order for that linear family. It is useful evidence that increasing the relevant fixed block parameter need not reduce its decay rate. The small-field ceiling must simultaneously satisfy the displayed M-dependent restriction. It does not identify the losses in the different nonlinear variational response (190), or equate differently named collar parameters across the papers.

The native reference and sharp-boundary result in Sections 6–7 uses the already admitted local NB12 reference. It does not use an unverified global metric summation.

8.2 The precise missing clauses

The remaining source-verification requirements are:

1. PRTII [5], equation (2.36) and its adjacent path-domain prose: admissible paths, all charged shells, treatment of the outer complement, endpoint blocks and any additive endpoint loss.
2. Its geometric assumptions (2.1)–(2.4), Lemma 2.1 and the summation clauses used by the original construction: the exact fixed collar and comparison factor κ must be matched.
3. The variational paper [6], Proposition 9 and the context of (189)–(190): the nonlinear decay loss, complex input domain, and dependence on the already chosen constants.

These exact original clauses are not verified here. The Wilson-source estimates [W] record several scale powers and rates but not the missing path convention. The background paper [4] is not a replacement for that absent definition. No statement here declares the native exterior free, nor silently charges it a new cost.

The geometric implication of [2] remains valid under its explicit hypotheses. If $s \geq 0$, $\rho_{\mathrm{eff}} > 0$, and top-to-depth-m points obey $d_{\mathrm{nat}}(x,y) \geq \rho_{\mathrm{eff}}(m-1)_+$ and $\sigma_y = L^{-m}$, then

$$\sigma_y^{-s} e^{-c d_{\mathrm{nat}}(x,y)} \leq L^s e^{-(c - s\log L/\rho_{\mathrm{eff}}) d_{\mathrm{nat}}(x,y)}. \tag{59}$$

For $s = 2$, a positive residual exponent needs $c\rho_{\mathrm{eff}} > 2\log L$. This implication is not an identification of the original c, ρ_{eff} or path metric. Uniform global shell counting additionally needs the exterior/path convention; depth crossing alone does not imply it.

8.3 A weighted-operator alternative and its Schur-source obstacle

A direct weighted operator estimate could avoid an absolute kernel summation. However, the constrained linear operator has the structural form

$$H = GQ^*(QGQ^*)^{-1}. \tag{60}$$

Here G is the corrected propagator and Q, B use the normalization of the background paper's (3.126). Its p. 422 distinguishes this from the original variational normalization, which inserts an additional assigned-scale factor; that factor must be restored before using the formula for a native target. For a top-supported target B in the displayed normalization, the fine source for G is $J_B = Q^*(QGQ^*)^{-1}B$, not Q^*B. The inverse Schur operator generally spreads the target over other shells. A valid application therefore requires a weighted bound on this complete source, for example an appropriately normalized estimate of the form

$$\left\| Q^*(QGQ^*)^{-1}B \right\|_{(0)} \leq C \left\| B \right\|_{\infty},$$

or a suitable weighted substitute, with any actual coordinate count retained. It also requires identifying the corresponding linearization of the nonlinear minimum.

The weighted global formula (3.47) of [4] is not used here as a verified displayed inequality, and no right-hand side is reconstructed from expectation. The exact weighted inequality, its use on the Schur source, and the nonlinear identification remain gates of this alternative route.

9 All-order factorial control of the bare contact sector

This section strengthens one remaining scope limitation of [2]. It controls growth with source order for *bare contact completion*; it does not replace the ordinary connected estimates in the preceding sections.

For the original finite-source $A = \sum_i \eta_i O_i$, $V = \sum_i \eta_i V_i$ and $t = \sum_i \eta_i t_i$, the exact conjugation is

$$e^{-D_V}(\rho e^A) = \rho e^B, \qquad B = e^V A - \int_0^1 e^{sV} t\, ds, \qquad \int \rho e^A F = \int \rho e^B e^V F. \tag{61}$$

At source order $r \geq 2$,

$$B_I = \frac{1}{(r-1)!} \sum_{\pi \in S(I)} V_{\pi_1} \cdots V_{\pi_{r-1}} O_{\pi_r} - \frac{1}{r!} \sum_{\pi \in S(I)} V_{\pi_1} \cdots V_{\pi_{r-1}} t_{\pi_r}. \tag{62}$$

The coefficient uses each original label once. All derivatives act on the coefficients to their right.

Theorem 9.1 (Factorial bare-contact bound)**.** *There is a constant C, independent of r, a, β, n and the supports, such that for $r \geq 2$, $u \geq 1$,*

$$\boxed{\|B_I\|_{L^u(\mu_\beta)} \leq u C^r r! a^{-2r} \min_{j \in I} \left\{ \|f_j\|_{\ell^1(\text{original anchors})} \prod_{i \neq j} \|f_i\|_\infty \right\}.} \tag{63}$$

The constant depends on the fixed group, stencils, original normalization and local-moment regime. Pointings and their averaging weights are retained. The ℓ^1 count uses original bond anchors; a separately counted color multiplicity is not counted twice.

Proof. An ordered s-fold invariant derivative of one plaquette action is a trace of its fixed four-link word with inserted bounded generators. Each differentiation acts on one of a fixed number of link occurrences. Thus

$$|X_{i_s} \cdots X_{i_1} s_p| \leq K_0 K_1^s, \tag{64}$$

and it vanishes unless each differentiated bond occurs in that plaquette. There is no factorial in (64). A first derivative has the stronger bound $|X_i s_p| \leq C\sqrt{s_p}$.

Expand the seed P or D into two plaquette first-derivative factors. After k applications of W, a term has $k+2$ action-derivative factors. The next W chooses one of those factors, a derivative coordinate on its plaquette, and one of the finitely many plaquettes in its new force coefficient. These choices total at most $C(k+2)$. Consequently $r-1$ applications create at most $C^r r!$ terms per seed anchor. This enumeration already includes the coordinate choices; multiplying by a separate connected-tree count would count them again.

For the scalar seed, use the literal Casimir identity

$$\ell[h] = C_F \sum_p h_p (N - s_p), \qquad h_p = \sum_{b \in \partial p} h_b.$$

At least one W acts when $r \geq 2$. It kills only the differentiated constant and creates two first-force factors. Every subsequent application preserves at least two such factors: it adds one and consumes at most one. All other action derivatives cost a further C^r, since their total order is at most $2r$. Two surviving forces are bounded by a sum of two original plaquette energies.

Fix any label j and its anchor b. Every new derivative anchor attaches to a plaquette already in the term, so the labeled anchor graph is connected and has diameter at most Cr in a fixed bounded-degree graph. The seed anchor lies in a ball with at most C^r possible vertices around b. Group the actual terms by this label-j anchor, retain $|f_j(b)|$, and bound the other weights by their sup norms. The seed possibilities add only an exponential factor. Periodic identifications cannot increase this graph-ball count.

The $r!$ permutations in the first sum of (62) are divided by $(r-1)!$, leaving only r; the second leaves one. These factors are absorbed by C^r. The physical coefficient is bounded by $C^r a^{-2r}(\beta+1)$ times the two original energies just described. Equation (3), Minkowski and the anchor count prove (63). Since j was arbitrary, take the minimum. □

The exact disconnected-contact vanishing of [2] also remains true. No ordinary spatial clustering is deduced from it. The proof applies to all r on any of the fixed compatible tori large enough for the original local stencils; repeated anchors and wrapping derivative graphs are allowed and do not enlarge the degree bound.

9.1 An independent analytic remainder estimate

For a single weight $\|f\|_\infty \le 1$, use the dimensionless parameter $\eta = \lambda c_\beta a^2$. Then

$$B(\lambda) = \lambda\beta e^{-\lambda W_f/3} G_f - \frac{\lambda}{3}\int_0^1 e^{-s\lambda W_f/3}(-\beta P[f] + \ell[f])\, ds, \tag{65}$$
$$R_f = B'(0) = \beta D[f] - \ell[f]/3.$$

Let Y contain all plaquettes incident to the actual moved-link set and all plaquettes in the original source/current coefficients, including their literal pointings. It is a closed determining set for this flow.

Proposition 9.2 (One-reserve analytic completion)**.** *There are $r_0, C > 0$, independent of volume, a and β, such that on every original real link configuration,*

$$|B(\lambda) - \lambda R_f| \le C|\lambda|^2(\beta+1)\sum_{p\in Y} s_p, \qquad |\lambda| \le r_0. \tag{66}$$

Proof. Complexify to $SL(N,\mathbb{C})$ and extend s_p holomorphically as $N - \frac{1}{2}\operatorname{tr}(U_p + U_p^{-1})$. A fixed tube about the compact group bounds link matrices and their inverses. The local vector field $-W_f/3$ and its local Lipschitz constant are bounded in the product maximum norm independently of the number of links. The ODE contraction argument therefore gives a holomorphic flow on a fixed complex-time disk, remaining in that tube. Traceless generators preserve determinant one.

Only the actual moved links vary. Put $q_p = U_p - I$ for $p \in Y$. Every force driving such a link is linear in the incident q_p and $q_p U_p^{-1}$ with bounded coefficients; this uses the zero trace of each generator. Differentiating the plaquette products gives $dq/d\lambda = \mathcal{A}(\lambda)q$, where the coefficient representation may depend on the nonlinear solution but has bounded row and column incidence. Its ℓ^2 operator norm is uniformly bounded. Gronwall along a complex radial segment gives

$$\sum_{p\in Y}\|q_p(\lambda)\|_{\mathrm{HS}}^2 \le C\sum_{p\in Y}\|q_p(0)\|_{\mathrm{HS}}^2 = 2C\sum_{p\in Y} s_p(U).$$

At the flowed field, $W_f G_f$, $W_f P[f]$ and $W_f \ell[f]$ still contain two first-force factors and bounded second action derivatives. Their absolute sums are therefore bounded by $C\sum_Y s_p(U)$. For the last expression use the holomorphic Casimir identity, retaining its constant until it is differentiated. Subtract the zero-time value of each term in (65) and integrate its flow derivative. This proves (66). □

Cauchy gives $r!C^r$ growth in the dimensionless source, independently confirming the factorial conclusion with a coarser union-support count. For real f, real compact-flow change of variables proves equality of source integrals. At each finite lattice, analytic continuation extends that identity locally to complex λ and complex test weights. No new contour is postulated.

Neither (66) nor (63) bounds the cumulants of the ordinary linear residual R_f. Integrating $e^{B(\lambda)}$ can still generate zeros near zero through that fluctuating part. The radius just proved scales as a^2 in the original physical source η. It is not a zero-free pressure disk or the fixed-source susceptibility theorem.

10 S94 and the continuum hierarchy: the remaining quantitative work

10.1 A native wall tail is not yet an ultraviolet scalar sum

The new reference-center comparison (58) controls one matched record. Complete S94 compares the entire source-zero construction on different physical tori, per physical volume. Besides a kernel ratio, it retains the changes of complete carriers, original coordinate dimensions, independent scalar increments, spatial actions, wrapped records and terminal terms.

There is a concrete summability issue with trying to use the native wall bound alone. For illustration, assume the usual weak ultraviolet scaling at a finer level j has $a_j = L^{-j}$ and $g_j^{-2} \asymp j$, and that the original cutoff has $R_j \asymp (\log g_j^{-2})^{p_0}$ with a fixed positive integer p_0. A bound charged separately to every current cell per physical volume would have the form

$$\begin{aligned} \mathfrak{m}_j &= L^{4j} g_j^{-r} (\log g_j^{-2})^s e^{-cR_j^2}, \\ \log \mathfrak{m}_j &= 4j \log L - cR_j^2 + \tfrac{r}{2} \log j + s \log\log j + O(1), \qquad R_j^2 \asymp (\log j)^{2p_0}. \end{aligned} \tag{67}$$

The linear term $4j \log L$ dominates the fixed power of $\log j$, so this majorant tends to infinity. Comparability inside the negative exponential has not been replaced by multiplicative comparability of the exponentials. This is a failure of that proposed bound, not a lower bound on the actual error. The original scalar comparison needs exact matching/cancellation, a spatial exit-distance gain, or an already justified saved scale factor. Superpolynomial decay in the inverse current coupling alone is insufficient after this cell count.

For a normalized complete kernel acting on 1, the derivative is exactly zero; such artificial reference changes should not be charged as errors. For a nonconstant inherited carrier, however, (50) keeps its derivative and covariance, and (56) keeps its independent scalar. The full comparison must use these identities before estimating.

10.2 Obligation ledger

Obligation	Advance in this note	Still required
Fixed-source H susceptibility	An extensive conditional sector is bounded at zero; its retained covariance is identified exactly.	A uniform bound on $\beta^2 n^{-4} \operatorname{Var}_{\mu_X}(K_{\mathcal{S}} G_H)$, or a verified multiscale contraction controlling the corresponding recursion.
O1: AE8/LM47	An explicit interior reference-center lift uses the original forest, pins, target product and Haar Jacobian.	General selected three-join prepared fields, anchored charts, simultaneous complex walls, original readout and old-domain identification.

Obligation	Advance in this note	Still required
O2: MT36	Actual conditional cumulants, effective-score jets and the complete degree-two current packet are bounded for a positive all-fields separated-link step. A native reference-center boundary response is bounded.	Identification and graded estimates for every inherited physical current, ordinary remainder, live-reference response and additional selector boundary in the actual mixed operations.
O3: scalar/contact filing	Factorial bare-contact growth; exact conditional covariance budget; all original absolute masses retained.	Nondegenerate interacting normalization and complete lower/contact filing through all transitions.
Metric gate	Common constants for the linear propagator family are checked from an additional primary source.	Exact exterior/path convention, fixed collar comparison, nonlinear decay losses and the complete native Schur-source estimate.
O4: S94	A same-record reference-center comparison and its raw/extracted scalar responses are explicit.	Cross-torus complete carriers and maps, independent absolute increments, volume-weighted scale summability, wrapped and terminal terms.
O5: continuum hierarchy	All-order growth is now controlled for one bare contact sector; no source order is silently identified with another.	A common cutoff-uniform hierarchy for all physical coefficients, compatible limits, distribution bounds, reflection positivity and Euclidean covariance after the actual prescriptions.
O6: Yang–Mills identification	The original nonlinear Wilson law and physical source normalization are preserved.	The final interacting theory and its local operator/dynamical identification; compactness of a smaller observable family alone does not supply it.

The concrete remaining susceptibility question is (18), with the actual local marginal (10). The complete one-step estimates make that a more structured problem than an unconstrained ordinary covariance. They do not yet supply the spatial or multiscale contraction needed to close it. No conditional implication in this note is presented as the completed Yang–Mills existence result.

References

[W] Wilson-source estimates, volumes 1–10 (lattice_theory/source_estimates in this collection), cited by source page. Chapters 10, 143, 155–159 and 161.

[2] *Bare physical contacts, susceptibility and native scales (bare_contacts.pdf).* Local Wilson moments, bare contact completion, source-averaged susceptibility and conditional native-scale geometry.

[3] T. Bałaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Commun. Math. Phys. **122** (1989), 355–392. DOI: 10.1007/BF01238433.

[4] T. Bałaban, *Propagators for lattice gauge theories in a background field*, Commun. Math. Phys. **99** (1985), 389–434. DOI: 10.1007/BF01240355. Relevant clauses are used especially from pp. 393–399 and 417–424. The global weighted formula (3.47) is explicitly not used as a verified displayed inequality.

[5] T. Bałaban, *Propagators and renormalization transformations for lattice gauge theories. II*, Commun. Math. Phys. **96** (1984), 223–250. DOI: 10.1007/BF01240221. Equation (2.36) and its original path-domain prose are not verified against the printed source here; they are listed as an unresolved source gate.

[6] T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Commun. Math. Phys. **102** (1985), 277–309. DOI: 10.1007/BF01229381. The missing parameter and domain context of (189)–(190) is not declared verified here.

[7] H. Shen, R. Zhu and X. Zhu, *A stochastic analysis approach to lattice Yang–Mills at strong coupling*, Commun. Math. Phys. **400** (2023), 805–851; arXiv:2204.12737. Its strong-coupling hypothesis is not substituted for the weak-coupling trajectory of this note.